Mar. 3

Chapter 6: Homomorphisms, Isomorphisms, and Cayley's Theorem

- **Def** Let $(G,*)$ and $(H,\star)$ be groups. A function $\varphi: G \to H$ is called a homomorphism
  if $\varphi(\underset{\text{in } G}{a*b}) = \varphi(a) \underset{\text{in } H}{\star} \varphi(b)$, $\forall a,b \in G$ $\quad$ φ respects the group operations

  "similar shape"

  - If, in addition, φ is bijective, then we say φ is an isomorphism

    "same shape"

- S_3 is isomorphic to D_3. We denote this $S_3 \cong D_3$

Ex. Consider $\langle (123) \rangle \leq S_3$ and $\langle r \rangle \leq D_3$

$\langle (123) \rangle$	id	(123)	(132)
id	e	(123)	(132)
(123)	(123)	(132)	e
(132)	(132)	e	(123)

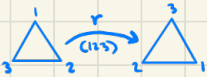
φ

$e \mapsto e$

$(123) \mapsto r$

$(132) \mapsto r^2$

$\langle r \rangle$	e	r	r^2
e	e	r	r^2
r	r	r^2	e
r^2	r^2	e	r



note: this depends on the bijection mapping

φ is an isomorphism. We can tell because the Cayley tables are the same

- **Exercise** construct an isomorphism between S_3 and D_3

  $S_3 = \langle (123), (23), \ldots \rangle$ $\quad$ $D_3 = \langle r, s : r^3 = s^2 = e, rs = sr^2 \rangle$

- There is no formula to construct isomorphisms every time, study the groups

- Let $G = \langle a \rangle$ with $|G| = n$ and $H = \langle b \rangle$ with $|H| = n$. Then $\varphi(a^k) = b^k$

  Want to show $\varphi(a^k \cdot a^\ell) = \varphi(a^k) \cdot \varphi(a^\ell)$

  $\varphi(a^{k+\ell}) = b^{k+\ell} = b^k \cdot b^\ell$

  here log means ln

- Particular homomorphism: $\log(a \cdot b) = \log(a) + \log(b)$

  $\ln(ab)$ asks $e^{\text{to what}} = ab$ $\quad$ $\psi: (\mathbb{R}^{>0}, \cdot) \to (\mathbb{R}, +)$ by $\psi(y) = \ln(y)$

  Converse, $e^{A+B} = e^A \cdot e^B$. Define $\varphi: (\mathbb{R}, +) \to (\mathbb{R}^{>0}, \cdot)$ by $\varphi(x) = e^x$

  exponentiation is an isomorphism from $(\mathbb{R}, +)$ to (positive $\mathbb{R}, \cdot$)

  φ is an isomorphism with inverse isomorphism ψ

★ this worksheet is the last of what may be on exam 1

Group Homomorphisms and Isomorphisms

Abstract Algebra

A function $\phi : G \to H$ between groups $(G, *)$ and $(H, \star)$ is a *homomorphism* if

$$\phi(a * b) = \phi(a) \star \phi(b) \quad \text{for all} \quad a, b \in G.$$

Loosely speaking, ϕ *respects or preserves the group operation.* From Wikipedia: The word homomorphism comes from the Ancient Greek language: homos meaning "same" and morphe meaning "form" or "shape". However, the word was apparently introduced to mathematics due to a (mis)translation and it should be "similar shape". If ϕ is also injective (one-to-one) and surjective (onto), then we say ϕ is an *isomorphism*. Iso is apparently Greek for 'equal.' In this case we think of G and H as being the exact same group.

1) Construct an isomorphism between $(\mathbb{Z}/2\mathbb{Z}, +)$ and $(S_2, \circ)$. Write out their group operation tables or Cayley tables to help verify your isomorphism. Is $(\{\pm 1\}, \cdot)$ also isomorphic to these groups?

$\mathbb{Z}/2\mathbb{Z} = \{\bar{0}, \bar{1}\}$, $S_2 = \{id, (12)\}$ $\qquad$ $(\{\pm 1\}, \cdot) = (\{1, -1\}, \cdot)$

$\mathbb{Z}/2\mathbb{Z}$	$\bar{0}$	$\bar{1}$
$\bar{0}$	$\bar{0}$	$\bar{1}$
$\bar{1}$	$\bar{1}$	$\bar{2}=\bar{0}$

S_2	id	(12)
id	id	(12)
(12)	(12)	id

$(\{\pm 1\}, \cdot)$	1	-1
1	1	-1
-1	-1	-1·-1=1

$\varphi : (\mathbb{Z}/2\mathbb{Z}, +) \hookrightarrow S_2$ given by $\varphi(\bar{0}) = id$, $\varphi(\bar{1}) = (12)$

$\varphi : (S_2, \circ) \hookrightarrow (\{\pm 1\}, \cdot)$: $\varphi(id) = 1$, $\varphi((12)) = -1$

$\varphi : (\mathbb{Z}/2\mathbb{Z}, +) \hookrightarrow (\{\pm 1\}, \cdot)$: $\varphi(\bar{0}) = 1$, $\varphi(\bar{1}) = -1$

So $(\{\pm 1\}, \cdot)$ is also isomorphic

2) Come up with two more examples of groups that look different at first glance, but are actually isomorphic. Ideas: Compute some multiplicative groups $\mathbb{Z}/p\mathbb{Z}^\times$ and compare to some additive groups $\mathbb{Z}/(p-1)\mathbb{Z}$. Also, D_3 and S_3. How about $\mathbb{Z}/m\mathbb{Z}$ and a cyclic subgroup of S_m?

$\mathbb{Z}/3\mathbb{Z}^\times = \{\bar{1}, \bar{2}\}$, $\mathbb{Z}/(3-1)\mathbb{Z} = \mathbb{Z}/2\mathbb{Z} = \{\bar{0}, \bar{1}\}$

$\mathbb{Z}/3\mathbb{Z}^\times$	$\bar{1}$	$\bar{2}$
$\bar{1}$	$\bar{1}$	$\bar{2}$
$\bar{2}$	$\bar{2}$	$\bar{2}\cdot\bar{2}=\bar{4}=\bar{1}$

$\mathbb{Z}/2\mathbb{Z}$	$\bar{0}$	$\bar{1}$
$\bar{0}$	$\bar{0}$	$\bar{1}$
$\bar{1}$	$\bar{1}$	$\bar{2}=\bar{0}$

$\varphi : \mathbb{Z}/3\mathbb{Z}^\times \hookrightarrow \mathbb{Z}/2\mathbb{Z}$

$\varphi(\bar{1}) = \bar{0}$

$\varphi(\bar{2}) = \bar{1}$

$\mathbb{Z}/2\mathbb{Z} = \{\bar{0}, \bar{1}\}$, $\langle (1\,2) \rangle \leq S_2$

here $(12)(12) = e$

$\mathbb{Z}/2\mathbb{Z}$	$\bar{0}$	$\bar{1}$
$\bar{0}$	$\bar{0}$	$\bar{1}$
$\bar{1}$	$\bar{1}$	$\bar{2}=\bar{0}$

$\langle (12) \rangle$	e	(12)
e	e	(12)
(12)	(12)	e

$\varphi : \mathbb{Z}/2\mathbb{Z} \hookrightarrow \langle (12) \rangle$

$\varphi(\bar{0}) = e$

$\varphi(\bar{1}) = (12)$

Claim: Suppose $\phi: G \to H$ is a hom. Then $\phi(e_G) = e_H$

Proof: $e_G * e_G = e_G$, so $\phi(e_G * e_G) = \phi(e_G)$. Since ϕ is a hom, $\phi(e_G * e_G) = \phi(e_G) * \phi(e_G)$.

Thus $\cancel{\phi(e_G)} * \phi(e_G) = \phi(e_G) = \cancel{\phi(e_G)} * e_H$. Thus $\phi(e_G) = e_H$.

group op. by $\phi(e_G)^{-1}$

3) Suppose $\phi : G \to H$ is an isomorphism. Prove $\phi(a^{-1}) = \phi(a)^{-1}$. for $\forall a \in G$

$a * a^{-1} = e_G$. Thus $\phi(a * a^{-1}) = \phi(e_G) = e_H$

$\phi(a) * \phi(a^{-1})$

Now $\phi(a) * \phi(a^{-1}) = e_H$. Group op. by $\phi(a)^{-1}$ on the left:

$\cancel{\phi(a)^{-1} * \phi(a)} * \phi(a^{-1}) = \phi(a)^{-1} * e_H$

$e_H * \phi(a^{-1}) = \phi(a)^{-1} * e_H$

$\phi(a^{-1}) = \phi(a)^{-1}$. $\square$

4) Suppose $\phi : G \to H$ is an isomorphism. Prove G is abelian if and only if H is abelian.

5) Suppose $\phi : G \to H$ is an isomorphism. If $A \leq G$, then prove $\phi(A) \leq H$. (Recall, $\leq$ means "is a subgroup of".)

6) Suppose $\phi : G \to H$ is an isomorphism. Prove $|a| = |\phi(a)|$.

7) Which of 3) through 6) remain true if we replace isomorphism with homomorphism?